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Explainable Home and Last-Mile Connectivity Diagnosis

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Abstract

NetSleuth AI diagnoses home internet issues by analyzing network measurements, identifying possible faults, and recommending useful tests and solutions.

Problem Description

Home internet issues are difficult to diagnose because different network problems can produce similar symptoms.

Key Issues:

Slow Internet: Unclear causes of reduced speed.

Latency & Packet Loss: Poor network performance.

Fault Identification: Difficulty locating the source of issues.

Limited Explanations: Basic tests provide little diagnostic insight.

Inefficient Troubleshooting: Repeated tests waste time.

Background

Home internet problems can originate from Wi-Fi, routers, DNS services, or the ISP's network. Traditional tools measure speed, latency, and packet loss but often cannot identify the exact cause using a single test.

Proposed Optimization

Combined Analysis: Analyze multiple network measurements to identify possible faults.
Explainable Diagnosis: Show the evidence supporting or contradicting each possible cause.
Intelligent Test Selection: Choose the next test that best distinguishes between possible faults.
Reliable Recommendations: Communicate uncertainty and suggest appropriate troubleshooting steps.

Literature & RFC Review

RFC 792 — ICMP: Defines network control messages used for reachability checks and error reporting. Supports basic connectivity diagnostics.
RFC 1035 — DNS: Specifies DNS implementation and queries. Supports measuring name-resolution delays and failures.
RFC 2680 — Packet Loss: Defines a standard metric for measuring packet loss across network paths. Supports loss analysis.
RFC 3393 — Delay Variation: Defines IP packet delay variation, commonly associated with jitter. Supports identifying inconsistent network performance.
Research Gap: These standards define individual measurements, but NetSleuth AI combines multiple observations, compares possible fault causes, and selects follow-up tests to improve explainability.

Proposed Solution

NetSleuth AI is an explainable network diagnosis system that identifies potential causes of home internet connectivity problems using real network measurements.
Data Collection: Measures gateway reachability, latency, packet loss, DNS response time, and route information where available.
Fault Analysis: Evaluates possible issues involving Wi-Fi, routers, DNS, and upstream network connectivity.
Evidence-Based Diagnosis: Compares measurements that support or contradict each possible cause.
Intelligent Test Selection: Selects the next useful test to distinguish between competing fault hypotheses.
Diagnostic Report: Presents likely causes, supporting evidence, uncertainty, and recommended troubleshooting actions.
Key Innovation: Instead of merely reporting poor connectivity, NetSleuth AI explains what the measurements suggest and what should be tested next.

Optimization Proposed

NetSleuth AI improves traditional network troubleshooting by combining multiple measurements, including latency, packet loss, DNS response time, and gateway reachability, to identify potential connectivity issues more accurately. Instead of relying on individual test results, it evaluates supporting and contradicting evidence for different fault hypotheses.

An intelligent next-test selection mechanism recommends additional tests based on the remaining uncertainties, helping distinguish between possible causes while reducing unnecessary testing. The system also generates explainable diagnostic reports containing likely causes, supporting evidence, and recommended actions. By communicating uncertainty and avoiding unsupported conclusions, NetSleuth AI aims to make network diagnosis more efficient, transparent, and reliable.

Solution Architecture & Design

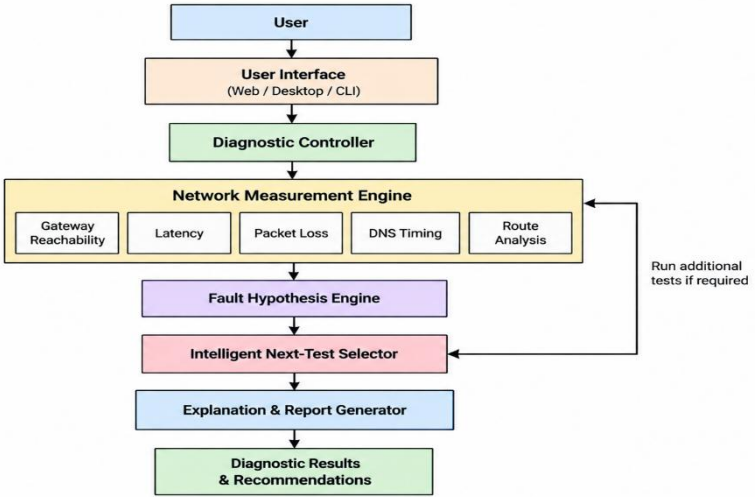


Figure: System Architecture of NetSleuth AI for Explainable Network Diagnosis

Data Flow Diagram

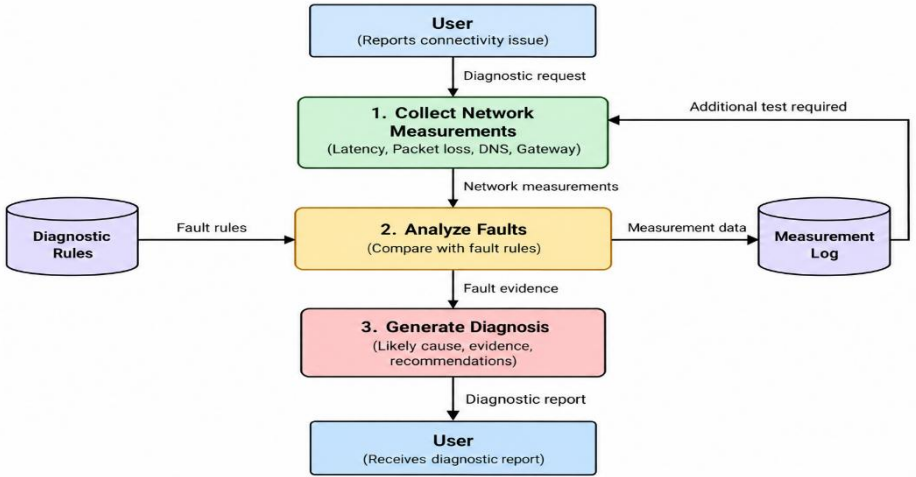


Figure: Data Flow Diagram (DFD) of NetSleuth AI

Deployment Timeline

The project will be developed in phases, from network measurement to testing and final deployment.
Phase 1 – Research: Study network diagnostics and identify common connectivity issues.
Phase 2 – Measurement: Implement latency, packet loss, DNS, and gateway reachability tests.
Phase 3 – Diagnosis: Develop the fault analysis engine and evidence-based explanations.
Phase 4 – Optimization: Implement intelligent next-test selection and diagnostic recommendations.
Phase 5 – Testing: Evaluate the system using controlled network fault scenarios.
Phase 6 – Deployment: Integrate all modules, prepare the dashboard, and demonstrate the final prototype.

Bibliography

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Thank You !